MALLESWARI GELLI

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CAREER OBJECTIVE:

MLOps Engineer and Data Scientist with 10+ years of experience in building and deploying AI/ML solutions. Expertise in developing enterprise-scale MLOps frameworks, cloud infrastructure, and automated ML pipelines. Proven track record in leading cross-functional teams and implementing data science platforms for R&D organizations.

EDUCATION:

- Masters certification in Data Science, iNeuron.ai (2022)
- Ph.D. Plant Breeding & Genetics, University of Nebraska, Lincoln. (2013)

CERTIFICATIONS:

- Data Science certification (Jul 2021-August 2022); Springboard Bootcamp
- MLOPs Foundations (Jan 2024 - April 2024); iNeuron
- Generative AI Masters (May 2024 - Dec 2024); Euron

TECHNICAL SKILLS

- **Databases:** snowflake, sqlite3, mysqlserver, mongodb, datalake.
- **AWS:** s3, serverless Lambda, Glue, Athena, EC2, EBS, EFS
- **ML/DL**: PyTorch, TensorFlow, scikit-learn, CNN, Computer Vision, NLP
- **Gen AI**: Large language model API: openai, huggingface, genimi, llama, frameworks: langchain, LCEL, lambda index, vector database: chroma, pinecone
- **MLOps & Cloud**: AWS SageMaker, Model Registry, MLflow, Docker, CI/CD (GitLab, GitHub Actions), Terraform
- **Tools & Frameworks**: Flask, FastAPI, Git, DVC, LangChain, Streamlit
- **Data Engineering:** Snowflake, Python, SQL, ETL pipelines, Feature Store

PROFESSSIONAL EXPERIENCE:

Sr. MLOPs Engineer; March 2024 - present (Syngenta, Remote)

- Architected enterprise-wide AI/ML Platform for Global Seeds R&D:
    - Implemented MLOps framework with SageMaker, MLflow, GitLab CI/CD, and automated monitoring.
    - Build essential infrastructure in collaboration with CloudOPs team.
    - Developed reusable SageMaker pipeline templates reducing deployment time by 40%.
    - In collaboration with data scientists, data engineers successfully implemented the data science platform to multiple R&D products and services like germplasm advancement, special adjustment, genomics, trait discovery and development.
- Engineered cost-effective, high-performance ML Infrastructure as code (IOC) Terraform:
    - Automated EC2 instance provisioning with GitLab CI/CD. Optimized storage with dynamic EBS volume attachment and AWS EFS integration.
    - Implemented for bioinformatics data science pipelines, achieved 30% cost savings and improved ML workload performance.

- Designed and deployed an automated Snowflake ETL pipeline for incremental data loading:
  - Implemented serverless AWS Lambda functions and SNS for real-time processing. Ensured efficient data synchronization and robust error handling.
  - Leveraged ETL pipeline template for multiple AI/ML projects across organization to load AI/ML model outputs from AWS s3 into snowflake schemas for downstream applications.
- Architected an in-house Snowflake feature store, serving as a central repository of features for data scientists and analysts:
  - Developed and deployed an automated template for creating feature store, entities and feature views.
  - Utilized the template for building feature store for various applications including genotyping, sequencing, phenotype characterization and spatial adjustment.
  - Facilitated feature reuse across multiple AI/ML projects, enhanced efficiency of data scientists by 40%
- Demonstrated strong leadership and communication skills:
  - Translated business requirements into technical solutions. Documented MLOps processes and workflows in confluence, and trained stakeholders, data scientists.
  - Effectively communicated complex technical concepts to non-technical stakeholders

Data Scientist/MLOPs Engineer; Dec 2023 – Feb 2024) (Johnston, Iowa)

MLOPs: Developed production-ready ML pipeline templates for diverse use cases

1. Customize, build and deploy end to end DL pipeline for computer vision: Image classification to accurately classify chest diseases from CT scan images.
   - Dataset Source - google drive (pipeline is built for smaller dataset)
   - Deploy Vgg16 model to AWS EC2 by Docker, CI/CD tool: Jenkins.
   - Integrated MLOPs tools: mlflow for experiment tracking and model management; DVC for data versioning, pipeline tracking
   - Built Flask web app [LinkedIn post][GitHub link]
2. Build and deploy an end to end ML classification pipeline for predicting if a US visa is approved or not based on certain features.
   - Data source: .csv files from MongoDB
   - Deploy classification ML model to AWS EC2 by Docker, CI/CD tool: GitHub Action
   - Integrated MLOPs tools: Evidently for checking data drift
   - Built FastAPI web app [LinkedIn post][GitHub link]
3. Build an end to end NLP pipeline to perform “named entity recognition (NER)” using a pre trained HuggingFace Transformer, BERT and deploy to Google Cloud Platform (GCP) with Docker, CI/CD tool: CircleCI [LinkedIn post][GitHub link]
4. Build End-to-End deep learning pipeline for a computer vision task: Object Detection with YOLOv7, deploy to AWS cloud with Docker, CI/CD GitHUb actions
   - Data source: Kaggle, data annotated and labeled using Roboflow.
   - Annotated data loaded into AWS s3 bucket [GitHub link]
5. By leveraging LLM, Build Virtual AI assistant application that inputs Audio and generates output Audio and deploy to AWS EC2 [GitHub Link]
6. Build end-to-end ML Regression pipeline for predicting student's performance and deploy Flask App to AWS Elastic Beanstack usingCI/CD tool: GitHub Actions[GitHub link]

7. Build end-to-end ML Regression pipeline for predicting housing price, deploy Flask app to cloud platform:Heroku with Docker, CI/CD tool: GitHub Actions[GitHub link]
    - Data source: .csv files from MongoDB; Data drift check with Evidently

Data Scientist; May 2022 – Nov 2023 (Calyxt, Inc., Roseville, Minnesota)

- Served as liaison, partnered with business and technology stakeholders to identify business needs, lead ideation sessions, build strategy, scope and implement analytical projects
- Build JIRA board to assist Scrum master to create EPICs, user stories and delegate tasks
- Web scraping of literature from Google scholar, PubMed and Wikipedia using Beautiful soup, html parser, Octoparse and SerpAPI.
- Developed NLP models to predict rank/class of abstracts (unstructured data) based on frequency of desired words,bag of words, TF-IDF reduced 20% of manual search time for researchers.
- Extracted, parsed data from plantcyc.org, mapped with schema and collaborated with system integration and data engineers to clean and store in database/Azure datalake.
- Build end to end machine learning, deep learning pipelines by implementing modular coding, custom logging and exception
- Collaborate with Software/Data Engineers to deploy AI/ML solutions into a live, production environment i.e Heroku, AWS, GCP for automation
- Built web application using Flask framework.
- Test and evaluate applications with different data sets, sources by training super users, collect feedback, contribute in developing enhancement workflows.

Senior Research Analyst; Sep 2017 – April 2022 (**Corteva Agriscience, Johnston, Iowa)**

- Lead cross functional team to develop & implement statistical/prediction models, tools to optimize training sets of breeding programs across crops, world for improved genetic gain.
- Served as portfolio manager: assist stakeholders defining the project scope and align (cross crop marker development projects) with organizational goals, finding cross functional resources, establish workplan, facilitate communicating project updates/challenges to stakeholders across all projects.
- Analyzed large structured, unstructured data i.e high-density multi-year, multi-locations field data and leverage NGS, high density plex genotyping data, haplotypes from multiple sources i.e databases (SQL, noSQL), URLs, APIs.
- Data wrangling/QC, mining, modeling, visualization with R (tidyverse) and python libraries (pandas, numpy, scipy, scikit-learn, matplotlib, seaborn etc), Spotfire, Power BI.
- Conduct genetic, statistical analysis for building genetic map, QTL mapping, GWAS, genomic prediction with supervised ML models (Linear, Logistic Regression, K-NN, Random Forrest, Decision Trees) to classify/predict the trait phenotype (disease resistance, yield)
- Built unsupervised machine learning models to cluster/group the germplasm with high density genotypes based on genetic similarity using hierarchical clustering, PCA, tSNE.
- Submit legal clearances, IP newly discovered QTLs and molecular markers.

Molecular Breeder/Geneticist; Nov 2013 – Sep 2017 (DuPont Pioneer, Johnson, Iowa)

- Served as product owner, oversee the development, validation and deployment of web apps/tools. Worked as a liaison between stakeholders, agile scrum and development (IT) teams by defining project scope, prioritization, building user stories, present development updates to clients and provide client feedback for enhancements. Contribute to prepare documentation and train end users/global breeders to enable marker selections.
- Performed lab experiments for validation of candidate genes to enable genomics-assisted breeding of native traits across crops (sorghum, maize and soybean)
- Conducted genetic, statistical analysis and test/validate new statistical methods, haplotype frame works for optimization, runtime efficiency for signal detection with varied density datasets, assist in making documentation and group trainings.
- Implemented NGS to sequence germplasm, identify SNP or Indel of interest.
- Perform marker design, testing and deployment to production of different chemistries (BHQ, TaqMan, KASPar) to enable high density genotyping for breeding selections.
- Performed phenotypic data analysis using SAS, R, python (experiment designs, statistical analysis, Chi square test, ANOVA, mixed effects model) and visualizations using SPOTFIRE and Tableu.
- Served as project manager, led projects by leveraging cross-functional expertise for mapping and validation of QTLs of multiple commercially important traits across sorghum, corn.